

Edge cases (an ambiguous routing decision) — Hints

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Only 2 hints — work through them in order, and don't jump ahead until you've genuinely tried. Each hint has 3 depth levels: **Basic** (the plain idea), **Intermediate** (proper LangGraph), **Advanced** (how a real supervisor system handles genuinely ambiguous cases). Read Basic first even if you already know LangGraph — it's the fastest way to spot exactly what each deeper level adds.

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Hint 1 — The idea, and the exact pieces

Basic Version

`supervisor_compare` built a supervisor for a task where the right order was never actually in question — research always comes before write. This exercise deliberately builds the opposite: two specialists whose jobs genuinely overlap, so the supervisor has a real decision to make, and can get it wrong.

Pick two specialists with descriptions that sound similar on purpose — for example a `research_agent` ("looks things up") and an `analysis_agent` ("figures out what data means"), where a prompt like "find out why sales dropped last quarter" could reasonably go to either one first.

Things to use:

- Two specialist node functions, each with a clearly written one-sentence description (this is what the supervisor's prompt will show the model).
- An LLM-based supervisor (like `supervisor_compare`'s Advanced version), since an `if`-based router can't make a genuinely ambiguous call at all.
- 5 test prompts, written to sit right on the boundary between the two specialists.
- A log that records which specialist got picked for each prompt.

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Intermediate Version

The two specialist descriptions are the entire experiment here — write them first, and write them so they genuinely overlap, not so one is obviously right:

- `research_agent` — "Looks up facts and data relevant to the task."
- `analysis_agent` — "Interprets data and explains what it means."

A prompt like "find out why sales dropped last quarter" needs both — you can't analyze what you haven't looked up, but "find out why" also sounds like an analysis request on its own. That's the genuine ambiguity, not a trick or a badly written description.

Look specifically at:

- **The supervisor's prompt needs both descriptions in it, verbatim** — the model is choosing between them based only on the text you give it, the same way `search_as_tool`'s Intermediate hint noted that a tool's docstring *is* its entire description to the model.
- **Running the same 5 prompts more than once** — a single run through an ambiguous case tells you what the model did *this time*; running each prompt 2-3 times shows whether the routing is consistent or genuinely unstable on the boundary cases.
- **Recording the raw routing decision, not just the final answer** — you need to see which specialist was picked, for which prompt, on which attempt, not just whether the final output looked reasonable.

Sketch the two specialist descriptions and your 5 test prompts before Hint 2.

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Advanced Version

Think about what "the supervisor routed correctly" even means once a prompt is genuinely ambiguous. For a prompt that could honestly go to either specialist first, there often isn't a single "correct" answer — the real question isn't "did it pick the right one," it's "was its pick consistent, and did the wrong pick still produce a recoverable result, or a silently broken one?"

This is where `Error propagation between agents` from Core Concepts applies directly: if the supervisor routes "find out why sales dropped" to `analysis_agent` first, and `analysis_agent` has nothing to analyze yet because no data was looked up, does it fail loudly (good — the supervisor can recover, maybe route to research next), or does it confidently analyze nothing and hand back a plausible-sounding but baseless answer (bad — the exact silent-failure trap Core Concepts warns about)?

The real design question isn't just "log which specialist got picked" — it's "what happens next, specifically, when the pick turns out to be the less useful one for that prompt?"

The extra pieces needed:

- A guard inside each specialist node that checks whether it actually has what it needs (`analysis_agent` checking for research data, for instance) and returns a clear "I need X first" signal instead of forging ahead on nothing.
- A supervisor that can read that signal and re-route — sending the task to the other specialist instead of treating a misroute as final.
- A written tally, after running all 5 prompts (2-3 times each), of exactly how often each specialist got picked first, and how often a misroute needed a second hop to recover.

Sketch this hardened version yourself before checking Hint 2.

Difference between Basic, Intermediate, and Advanced: Basic names the two overlapping specialists and the exact pieces (LLM-based router, 5 boundary prompts, a log) for a first working ambiguous-routing setup. Intermediate writes real, genuinely-overlapping descriptions and adds repeated runs to check consistency, not just a single pass. Advanced asks the harder question underneath "which one got picked" — what happens when the pick was the less useful one — and adds a guard-and-recover mechanism, which is the same "don't let a bad handoff silently produce a confident wrong answer" idea Core Concepts' Error propagation section names directly.

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Hint 2 — The plan, and almost the whole thing

Basic Version

```
research_agent: "Looks up facts and data relevant to the task."
analysis_agent: "Interprets data and explains what it means."

supervisor_node:
    ask the model to pick research_agent or analysis_agent for the task,
    given both descriptions
    log the task and the pick

5 test prompts, on purpose ambiguous, for example:
    "find out why sales dropped last quarter"
    "what does this survey data actually tell us"
    "look into the causes of the outage"
    "explain what happened to our conversion rate"
    "dig into the reasons behind the delay"

run all 5 through the supervisor, print the routing log
```

Expected output if you run just this (nothing calls the graph yet): nothing — a graph isn't invoked until you call `graph.invoke(...)` with a real prompt. Add a loop over the 5 prompts, invoking the graph for each one, to see the routing log fill in.

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Intermediate Version

```
from typing import TypedDict
from langgraph.graph import StateGraph, START, END
from langgraph.types import Command

SPECIALISTS = {
    "research_agent": "Looks up facts and data relevant to the task.",
    "analysis_agent": "Interprets data and explains what it means.",
}

class RoutingState(TypedDict):
    task: str
    picked: str

def supervisor(state: RoutingState) -> Command:
    descriptions = "\n".join(f"- {name}: {desc}" for name, desc in SPECIALISTS.items())
    prompt = (
        f"Task: {state['task']}\n\n"
        f"Choose exactly one specialist to handle this first:\n{descriptions}\n\n"
        "Reply with only the specialist's name."
    )
    response = model.invoke(prompt)
    picked = response.content.strip()
    return Command(update={"picked": picked}, goto=END)

builder = StateGraph(RoutingState)
builder.add_node("supervisor", supervisor)
builder.add_edge(START, "supervisor")
graph = builder.compile()

test_prompts = [
    "find out why sales dropped last quarter",
    "what does this survey data actually tell us",
    "look into the causes of the outage",
    "explain what happened to our conversion rate",
    "dig into the reasons behind the delay",
]
```

```
routing_log = []
for prompt in test_prompts:
    result = graph.invoke({"task": prompt, "picked": ""})
    routing_log.append((prompt, result["picked"]))
    print(f"{prompt!r} -> {result['picked']}")
```

Run this once, then run it again — notice whether the same prompt gets routed the same way both times — before checking Advanced.

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Advanced Version

```

each specialist checks what it actually has before doing work:
  analysis_agent: if no research data in state -> return a
                  "needs_research_first" signal instead of forging ahead

supervisor reads that signal:
  if a specialist says it needs the other one first -> route there, then
  come back -- log this as a "recovered misroute", not a final answer

run all 5 prompts, 3 times each -> tally:
  how often research_agent was picked first
  how often analysis_agent was picked first
  how often a misroute needed a second hop to recover

```

Here's almost the whole thing — fill in the missing piece yourself:

```

class RoutingState(TypedDict):
    task: str
    picked: str
    research_data: str
    misrouted: bool

def analysis_agent(state: RoutingState) -> Command:
    if not state.get("research_data"):
        # this specialist was picked, but has nothing to analyze yet
        return Command(
            update={"misrouted": True},
            goto="research_agent",
        )
    # ... do the real analysis here ...
    return Command(update={}, goto=END)

def research_agent(state: RoutingState) -> Command:
    # your turn: fill in research_data, then decide where to go next --
    # if this run started as a misroute (state["misrouted"]), it should
    # go back to analysis_agent now that research_data exists
    ...

```

Fill in `research_agent` yourself, run the full tally across all 5 prompts x 3 attempts, then compare your written conclusion against the [Solution](#).

Difference between Basic, Intermediate, and Advanced: the same underlying shape (2 overlapping specialist descriptions, an LLM decides, a log records the pick) at 3 completeness levels — Basic sketches the prompts and confirms nothing runs until invoked, Intermediate is a complete, working router you can run repeatedly to check consistency, and Advanced adds the guard-and-recover mechanism that turns a misroute from a silent wrong answer into a visible, logged, recoverable second hop.

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Full solution: [Show me the solution](#)